

How Can **Gene Drives** help farmers grow more fresh berries?

The Problem:

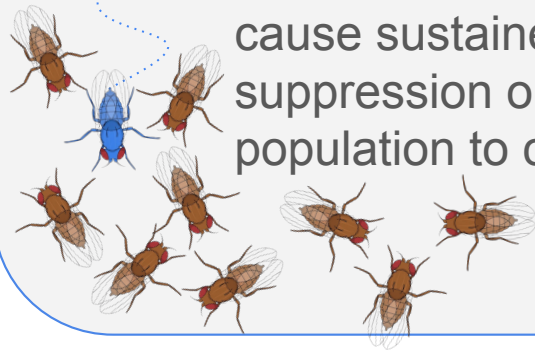
Blueberry growers in maine were estimated to lose up to **\$6.8 million per year** in 2020 due to invasive fruit flies [1].

- These invasive fruit flies can **infest ripe fruit** in contrast to their native
- Effect many fruits and berries like **blueberries, cherries, grapes, plums**, etc
- counterparts that only infest rotten fruit
- Current **solutions are becoming less effective** and can harm other native species

The Solution:

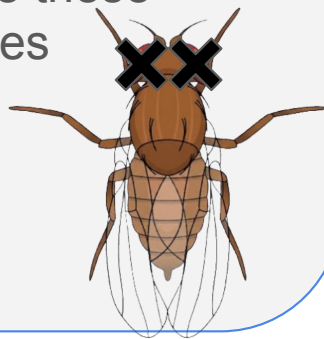
- **Gene drives**

- Release genetically engineered “trojan horse” flies that can cause sustained suppression or the population to crash



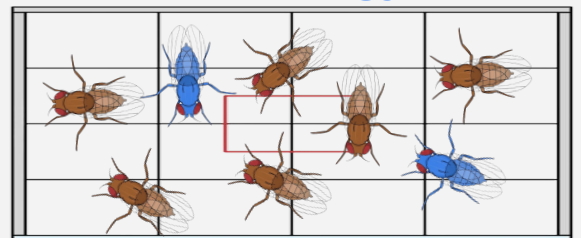
Results:

- **Gene drives** have been shown in labs to effectively suppress and in some cases eradicate these invasive fruit flies



Request:

- **Support controlled field tests** in your area and become the **first** to implement this **powerful new technology!**



For more information please reach out to Casey O'Brien at cpobrie6@ncsu.edu

[1] D Adeline Yeh, Francis A Drummond, Miguel I Gómez, Xiaoli Fan, The Economic Impacts and Management of Spotted Wing Drosophila (*Drosophila Suzukii*): The Case of Wild Blueberries in Maine, *Journal of Economic Entomology*, Volume 113, Issue 3, June 2020, Pages 1262–1269, <https://doi.org/10.1093/jee/toz360>